# Варианты лабораторной работы №1

Функция для многочленов Лагранжа, Ньютона с разделёнными разностями

| № варианта | Таблица значений функции | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | $x_i$ | 0,32 | 0,37 | 0,41 | 0,45 | 0,61 | 0,65 |
| | $y_i$ | 1,4126 | 2,521 | 4,102 | 3,110 | 3,098 | 2,711 |
| 2. | $x_i$ | −0,12 | −0,01 | 0,12 | 1,44 | 2,00 | 3,69 |
| | $y_i$ | 4,0012 | 4,2381 | 7,4962 | 13,2390 | 14,0907 | 15,4123 |
| 3. | $x_i$ | −0,65 | −0,51 | −0,43 | −0,31 | −0,15 | −0,01 |
| | $y_i$ | 13,651 | 14,877 | 15,764 | 19,112 | 21,000 | 23,578 |
| 4. | $x_i$ | −5,343 | −5,289 | −4,469 | −4,128 | −3,769 | −3,089 |
| | $y_i$ | −8,923 | −11,650 | −6,348 | −3,714 | −5,324 | −1,789 |
| 5. | $x_i$ | 1,41 | 2,31 | 4,13 | 4,89 | 5,31 | 6,01 |
| | $y_i$ | −1,4156 | 2,3901 | 3,0567 | 0,9812 | 4,1345 | 2,7569 |
| 6. | $x_i$ | −13,15 | −12,57 | −10,13 | −9,86 | −8,05 | −6,90 |
| | $y_i$ | 0,128 | −1,901 | 3,341 | 5,378 | 4,692 | 6,841 |
| 7. | $x_i$ | 3,99 | 4,15 | 4,25 | 4,62 | 5,12 | 5,74 |
| | $y_i$ | 5,8901 | 8,1230 | 9,6782 | 14,4591 | 12,9821 | 11,5329 |
| 8. | $x_i$ | −2,124 | −2,057 | −1,987 | −1,799 | −1,667 | −1,552 |
| | $y_i$ | 31,897 | 40,783 | 41,971 | 46,414 | 47,317 | 49,124 |
| 9. | $x_i$ | 3,50 | 3,67 | 3,80 | 3,91 | 4,75 | 5,42 |
| | $y_i$ | −19,567 | −17,849 | −13,681 | −11,831 | −9,451 | −10,931 |
| 10. | $x_i$ | 2,67 | 2,74 | 2,79 | 2,83 | 2,91 | 3,03 |
| | $y_i$ | 3,5678 | 3,3289 | 3,5076 | 3,9065 | 4,9076 | 5,5327 |
| 11. | $x_i$ | −9,15 | −8,79 | −8,43 | −8,15 | −7,45 | −7,35 |
| | $y_i$ | −2,677 | −0,990 | −1,666 | 0,433 | 2,588 | 3,999 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 12. | $x_i$ | 8,15 | 8,41 | 8,99 | 9,07 | 9,43 | 9,68 | |
| | $y_i$ | −10,7688 | −9,4533 | −8,1818 | −3,3311 | −2,4498 | −2,1111 | |
| 13. | $x_i$ | −4,12 | −3,87 | −3,60 | −3,12 | −2,78 | −2,23 | |
| | $y_i$ | 6,222 | 9,330 | 10,129 | 12,341 | 15,834 | 13,657 | |
| 14. | $x_i$ | −1,93 | −1,46 | −0,87 | −0,53 | −0,29 | −0,05 | |
| | $y_i$ | 18,3456 | 19,9472 | 21,91492 | 21,8492 | 23,1236 | 25,8484 | |
| 15. | $x_i$ | 11,153 | 11,454 | 11,673 | 11,879 | 12,009 | 12,231 | 12,549 |
| | $y_i$ | −3,234 | 5,321 | −1,123 | 0,393 | 8,939 | 141,231 | 15,001 |
| 16. | $x_i$ | −8,81 | −8,19 | −7,45 | −6,57 | −5,89 | −4,13 | −3,67 |
| | $y_i$ | 5,3722 | 4,6237 | 3,1234 | 8,4568 | 11,0003 | 17,7384 | 22,9283 |
| 17. | $x_i$ | 9,49 | 9,57 | 9,73 | 9,89 | 10,11 | 10,45 | 10,83 |
| | $y_i$ | −15,123 | −13,999 | −13,877 | −13,341 | −12,003 | −12,939 | −12,009 |
| 18. | $x_i$ | −11,02 | −10,78 | −10,23 | −9,89 | −9,65 | −9,03 | |
| | $y_i$ | 21,1230 | 22,8279 | 25,0046 | 27,5928 | 28,9933 | 29,1933 | |
| 19. | $x_i$ | −0,01 | 0,21 | 0,35 | 0,67 | 0,89 | 1,17 | |
| | $y_i$ | 16,12 | 20,83 | 25,93 | 26,92 | 27,92 | 30,11 | |
| 20. | $x_i$ | 5,15 | 5,50 | 6,11 | 6,37 | 7,01 | 7,15 | 7,94 |
| | $y_i$ | 1,6782 | 2,9882 | 3,3030 | 3,9929 | 6,9243 | 8,9393 | 10,3933 |
| 21. | $x_i$ | −9,12 | −8,78 | −8,34 | −8,09 | −7,86 | −7,52 | |
| | $y_i$ | 21,123 | 24,323 | 26,545 | 30,883 | 28,993 | 32,323 | |
| 22. | $x_i$ | 0,32 | 0,37 | 0,41 | 0,45 | 0,61 | 0,65 | |
| | $y_i$ | 1,4126 | 2,521 | 4,102 | 3,110 | 3,098 | 2,711 | |
| 23. | $x_i$ | −0,12 | −0,01 | 0,12 | 1,44 | 2,00 | 3,69 | |
| | $y_i$ | 4,0012 | 4,2381 | 7,4962 | 13,2390 | 14,0907 | 15,4123 | |
| 24. | $x_i$ | −0,65 | −0,51 | −0,43 | −0,31 | −0,15 | −0,01 | |
| | $y_i$ | 13,651 | 14,877 | 15,764 | 19,112 | 21,000 | 23,578 | |
| 25. | $x_i$ | −5,343 | −5,289 | −4,469 | −4,128 | −3,769 | −3,089 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $y_i$ | −8,923 | −11,650 | −6,348 | −3,714 | −5,324 | −1,789 | |
| 26. | $x_i$ | 1,41 | 2,31 | 4,13 | 4,89 | 5,31 | 6,01 | |
| | $y_i$ | −1,4156 | 2,3901 | 3,0567 | 0,9812 | 4,1345 | 2,7569 | |
| 27. | $x_i$ | −13,15 | −12,57 | −10,13 | −9,86 | −8,05 | −6,90 | |
| | $y_i$ | 0,128 | −1,901 | 3,341 | 5,378 | 4,692 | 6,841 | |
| 28. | $x_i$ | 3,99 | 4,15 | 4,25 | 4,62 | 5,12 | 5,74 | |
| | $y_i$ | 5,8901 | 8,1230 | 9,6782 | 14,4591 | 12,9821 | 11,5329 | |
| 29. | $x_i$ | −2,124 | −2,057 | −1,987 | −1,799 | −1,667 | −1,552 | |
| | $y_i$ | 31,897 | 40,783 | 41,971 | 46,414 | 47,317 | 49,124 | |
| 30. | $x_i$ | 3,50 | 3,67 | 3,80 | 3,91 | 4,75 | 5,42 | |
| | $y_i$ | −19,567 | −17,849 | −13,681 | −11,831 | −9,451 | −10,931 | |
| 31. | $x_i$ | 2,67 | 2,74 | 2,79 | 2,83 | 2,91 | 3,03 | |
| | $y_i$ | 3,5678 | 3,3289 | 3,5076 | 3,9065 | 4,9076 | 5,5327 | |
| 32. | $x_i$ | −9,15 | −8,79 | −8,43 | −8,15 | −7,45 | −7,35 | |
| | $y_i$ | −2,677 | −0,990 | −1,666 | 0,433 | 2,588 | 3,999 | |
| 33. | $x_i$ | 8,15 | 8,41 | 8,99 | 9,07 | 9,43 | 9,68 | |
| | $y_i$ | −10,7688 | −9,4533 | −8,1818 | −3,3311 | −2,4498 | −2,1111 | |
| 34. | $x_i$ | −4,12 | −3,87 | −3,60 | −3,12 | −2,78 | −2,23 | |
| | $y_i$ | 6,222 | 9,330 | 10,129 | 12,341 | 15,834 | 13,657 | |
| 35. | $x_i$ | −1,93 | −1,46 | −0,87 | −0,53 | −0,29 | −0,05 | |
| | $y_i$ | 18,3456 | 19,9472 | 21,91492 | 21,8492 | 23,1236 | 25,8484 | |
| 36. | $x_i$ | 11,153 | 11,454 | 11,673 | 11,879 | 12,009 | 12,231 | 12,549 |
| | $y_i$ | −3,234 | 5,321 | −1,123 | 0,393 | 8,939 | 141,231 | 15,001 |
| 37. | $x_i$ | −8,81 | −8,19 | −7,45 | −6,57 | −5,89 | −4,13 | −3,67 |
| | $y_i$ | 5,3722 | 4,6237 | 3,1234 | 8,4568 | 11,0003 | 17,7384 | 22,9283 |
| 38. | $x_i$ | 9,49 | 9,57 | 9,73 | 9,89 | 10,11 | 10,45 | 10,83 |
| | $y_i$ | −15,123 | −13,999 | −13,877 | −13,341 | −12,003 | −12,939 | −12,009 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 39. | $x_i$ | −11,02 | −10,78 | −10,23 | −9,89 | −9,65 | −9,03 | |
| | $y_i$ | 21,1230 | 22,8279 | 25,0046 | 27,5928 | 28,9933 | 29,1933 | |
| 40. | $x_i$ | −0,01 | 0,21 | 0,35 | 0,67 | 0,89 | 1,17 | |
| | $y_i$ | 16,12 | 20,83 | 25,93 | 26,92 | 27,92 | 30,11 | |
| 41. | $x_i$ | 5,15 | 5,50 | 6,11 | 6,37 | 7,01 | 7,15 | 7,94 |
| | $y_i$ | 1,6782 | 2,9882 | 3,3030 | 3,9929 | 6,9243 | 8,9393 | 10,3933 |
| 42. | $x_i$ | −9,12 | −8,78 | −8,34 | −8,09 | −7,86 | −7,52 | |
| | $y_i$ | 21,123 | 24,323 | 26,545 | 30,883 | 28,993 | 32,323 | |
| 43. | $x_i$ | 0,32 | 0,37 | 0,41 | 0,45 | 0,61 | 0,65 | |
| | $y_i$ | 1,4126 | 2,521 | 4,102 | 3,110 | 3,098 | 2,711 | |
| 44. | $x_i$ | −0,12 | −0,01 | 0,12 | 1,44 | 2,00 | 3,69 | |
| | $y_i$ | 4,0012 | 4,2381 | 7,4962 | 13,2390 | 14,0907 | 15,4123 | |
| 45. | $x_i$ | −0,65 | −0,51 | −0,43 | −0,31 | −0,15 | −0,01 | |
| | $y_i$ | 13,651 | 14,877 | 15,764 | 19,112 | 21,000 | 23,578 | |
| 46. | $x_i$ | −5,343 | −5,289 | −4,469 | −4,128 | −3,769 | −3,089 | |
| | $y_i$ | −8,923 | −11,650 | −6,348 | −3,714 | −5,324 | −1,789 | |
| 47. | $x_i$ | 1,41 | 2,31 | 4,13 | 4,89 | 5,31 | 6,01 | |
| | $y_i$ | −1,4156 | 2,3901 | 3,0567 | 0,9812 | 4,1345 | 2,7569 | |
| 48. | $x_i$ | −13,15 | −12,57 | −10,13 | −9,86 | −8,05 | −6,90 | |
| | $y_i$ | 0,128 | −1,901 | 3,341 | 5,378 | 4,692 | 6,841 | |
| 49. | $x_i$ | 3,99 | 4,15 | 4,25 | 4,62 | 5,12 | 5,74 | |
| | $y_i$ | 5,8901 | 8,1230 | 9,6782 | 14,4591 | 12,9821 | 11,5329 | |
| 50. | $x_i$ | −2,124 | −2,057 | −1,987 | −1,799 | −1,667 | −1,552 | |
| | $y_i$ | 31,897 | 40,783 | 41,971 | 46,414 | 47,317 | 49,124 | |
| 51. | $x_i$ | 3,50 | 3,67 | 3,80 | 3,91 | 4,75 | 5,42 | |
| | $y_i$ | −19,567 | −17,849 | −13,681 | −11,831 | −9,451 | −10,931 | |
| 52. | $x_i$ | 2,67 | 2,74 | 2,79 | 2,83 | 2,91 | 3,03 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $y_i$ | 3,5678 | 3,3289 | 3,5076 | 3,9065 | 4,9076 | 5,5327 | |
| 53. | $x_i$ | −9,15 | −8,79 | −8,43 | −8,15 | −7,45 | −7,35 | |
| | $y_i$ | −2,677 | −0,990 | −1,666 | 0,433 | 2,588 | 3,999 | |
| 54. | $x_i$ | 8,15 | 8,41 | 8,99 | 9,07 | 9,43 | 9,68 | |
| | $y_i$ | −10,7688 | −9,4533 | −8,1818 | −3,3311 | −2,4498 | −2,1111 | |
| 55. | $x_i$ | −4,12 | −3,87 | −3,60 | −3,12 | −2,78 | −2,23 | |
| | $y_i$ | 6,222 | 9,330 | 10,129 | 12,341 | 15,834 | 13,657 | |
| 56. | $x_i$ | −1,93 | −1,46 | −0,87 | −0,53 | −0,29 | −0,05 | |
| | $y_i$ | 18,3456 | 19,9472 | 21,91492 | 21,8492 | 23,1236 | 25,8484 | |
| 57. | $x_i$ | 11,153 | 11,454 | 11,673 | 11,879 | 12,009 | 12,231 | 12,549 |
| | $y_i$ | −3,234 | 5,321 | −1,123 | 0,393 | 8,939 | 141,231 | 15,001 |
| 58. | $x_i$ | −8,81 | −8,19 | −7,45 | −6,57 | −5,89 | −4,13 | −3,67 |
| | $y_i$ | 5,3722 | 4,6237 | 3,1234 | 8,4568 | 11,0003 | 17,7384 | 22,9283 |
| 59. | $x_i$ | 9,49 | 9,57 | 9,73 | 9,89 | 10,11 | 10,45 | 10,83 |
| | $y_i$ | −15,123 | −13,999 | −13,877 | −13,341 | −12,003 | −12,939 | −12,009 |
| 60. | $x_i$ | −11,02 | −10,78 | −10,23 | −9,89 | −9,65 | −9,03 | |
| | $y_i$ | 21,1230 | 22,8279 | 25,0046 | 27,5928 | 28,9933 | 29,1933 | |
| 61. | $x_i$ | −0,01 | 0,21 | 0,35 | 0,67 | 0,89 | 1,17 | |
| | $y_i$ | 16,12 | 20,83 | 25,93 | 26,92 | 27,92 | 30,11 | |
| 62. | $x_i$ | 5,15 | 5,50 | 6,11 | 6,37 | 7,01 | 7,15 | 7,94 |
| | $y_i$ | 1,6782 | 2,9882 | 3,3030 | 3,9929 | 6,9243 | 8,9393 | 10,3933 |
| 63. | $x_i$ | −9,12 | −8,78 | −8,34 | −8,09 | −7,86 | −7,52 | |
| | $y_i$ | 21,123 | 24,323 | 26,545 | 30,883 | 28,993 | 32,323 | |
| 64. | $x_i$ | 0,32 | 0,37 | 0,41 | 0,45 | 0,61 | 0,65 | |
| | $y_i$ | 1,4126 | 2,521 | 4,102 | 3,110 | 3,098 | 2,711 | |
| 65. | $x_i$ | −0,12 | −0,01 | 0,12 | 1,44 | 2,00 | 3,69 | |
| | $y_i$ | 4,0012 | 4,2381 | 7,4962 | 13,2390 | 14,0907 | 15,4123 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 66. | $x_i$ | $-0{,}65$ | $-0{,}51$ | $-0{,}43$ | $-0{,}31$ | $-0{,}15$ | $-0{,}01$ |
| | $y_i$ | 13,651 | 14,877 | 15,764 | 19,112 | 21,000 | 23,578 |
| 67. | $x_i$ | $-5{,}343$ | $-5{,}289$ | $-4{,}469$ | $-4{,}128$ | $-3{,}769$ | $-3{,}089$ |
| | $y_i$ | $-8{,}923$ | $-11{,}650$ | $-6{,}348$ | $-3{,}714$ | $-5{,}324$ | $-1{,}789$ |
| 68. | $x_i$ | 1,41 | 2,31 | 4,13 | 4,89 | 5,31 | 6,01 |
| | $y_i$ | $-1{,}4156$ | 2,3901 | 3,0567 | 0,9812 | 4,1345 | 2,7569 |
| 69. | $x_i$ | $-13{,}15$ | $-12{,}57$ | $-10{,}13$ | $-9{,}86$ | $-8{,}05$ | $-6{,}90$ |
| | $y_i$ | 0,128 | $-1{,}901$ | 3,341 | 5,378 | 4,692 | 6,841 |
| 70. | $x_i$ | 3,99 | 4,15 | 4,25 | 4,62 | 5,12 | 5,74 |
| | $y_i$ | 5,8901 | 8,1230 | 9,6782 | 14,4591 | 12,9821 | 11,5329 |
| 71. | $x_i$ | $-2{,}124$ | $-2{,}057$ | $-1{,}987$ | $-1{,}799$ | $-1{,}667$ | $-1{,}552$ |
| | $y_i$ | 31,897 | 40,783 | 41,971 | 46,414 | 47,317 | 49,124 |
| 72. | $x_i$ | 3,50 | 3,67 | 3,80 | 3,91 | 4,75 | 5,42 |
| | $y_i$ | $-19{,}567$ | $-17{,}849$ | $-13{,}681$ | $-11{,}831$ | $-9{,}451$ | $-10{,}931$ |
| 73. | $x_i$ | 2,67 | 2,74 | 2,79 | 2,83 | 2,91 | 3,03 |
| | $y_i$ | 3,5678 | 3,3289 | 3,5076 | 3,9065 | 4,9076 | 5,5327 |
| 74. | $x_i$ | $-9{,}15$ | $-8{,}79$ | $-8{,}43$ | $-8{,}15$ | $-7{,}45$ | $-7{,}35$ |
| | $y_i$ | $-2{,}677$ | $-0{,}990$ | $-1{,}666$ | 0,433 | 2,588 | 3,999 |
| 75. | $x_i$ | 8,15 | 8,41 | 8,99 | 9,07 | 9,43 | 9,68 |
| | $y_i$ | $-10{,}7688$ | $-9{,}4533$ | $-8{,}1818$ | $-3{,}3311$ | $-2{,}4498$ | $-2{,}1111$ |

Точки интерполяции (экстраполяции) многочленами Лагранжа, Ньютона с разделёнными разностями

| № варианта | Многочлен Лагранжа, схема Эйткена | Многочлен Ньютона с разделёнными разностями |
|---|---|---|
| 1. | $x = 0{,}39$ | $x^{(1)} = 0{,}66$<br>$x^{(2)} = 0{,}30$ |

| | | |
|---|---|---|
| 2. | $x = -0{,}09$ | $x^{(1)} = -0{,}11$<br>$x^{(2)} = 1{,}34$ |
| 3. | $x = -0{,}59$ | $x^{(1)} = -0{,}18$<br>$x^{(2)} = -0{,}47$ |
| 4. | $x = -3{,}941$ | $x^{(1)} = -5{,}687$<br>$x^{(2)} = -3{,}244$ |
| 5. | $x = 3{,}61$ | $x^{(1)} = 2{,}78$<br>$x^{(2)} = 5{,}19$ |
| 6. | $x = -7{,}18$ | $x^{(1)} = -13{,}48$<br>$x^{(2)} = -7{,}50$ |
| 7. | $x = 5{,}62$ | $x^{(1)} = 4{,}19$<br>$x^{(2)} = 5{,}45$ |
| 8. | $x = -1{,}498$ | $x^{(1)} = -1{,}744$<br>$x^{(2)} = -2{,}008$ |
| 9. | $x = 3{,}85$ | $x^{(1)} = 4{,}88$<br>$x^{(2)} = 3{,}57$ |
| 10. | $x = 2{,}81$ | $x^{(1)} = 2{,}75$<br>$x^{(2)} = 2{,}86$ |
| 11. | $x = -7{,}42$ | $x^{(1)} = -8{,}92$<br>$x^{(2)} = -7{,}39$ |
| 12. | $x = 7{,}48$ | $x^{(1)} = 8{,}75$<br>$x^{(2)} = 9{,}29$ |
| 13. | $x = -3{,}45$ | $x^{(1)} = -3{,}95$<br>$x^{(2)} = -2{,}51$ |
| 14. | $x = -0{,}15$ | $x^{(1)} = -1{,}77$<br>$x^{(2)} = -0{,}15$ |
| 15. | $x = 12{,}776$ | $x^{(1)} = 11{,}515$<br>$x^{(2)} = 11{,}995$ |
| 16. | $x = -8{,}91$ | $x^{(1)} = -3{,}81$<br>$x^{(2)} = -8{,}52$ |

| | | |
|---|---|---|
| 17. | $x = 9{,}76$ | $x^{(1)} = 10{,}32$<br>$x^{(2)} = 9{,}66$ |
| 18. | $x = -9{,}22$ | $x^{(1)} = -10{,}44$<br>$x^{(2)} = -9{,}41$ |
| 19. | $x = -0{,}16$ | $x^{(1)} = 0{,}27$<br>$x^{(2)} = 0{,}72$ |
| 20. | $x = 8{,}02$ | $x^{(1)} = 5{,}32$<br>$x^{(2)} = 7{,}09$ |
| 21. | $x = -7{,}43$ | $x^{(1)} = -7{,}61$<br>$x^{(2)} = -8{,}87$ |
| 22. | $x = 0{,}44$ | $x^{(1)} = 0{,}33$<br>$x^{(2)} = 0{,}68$ |
| 23. | $x = 0{,}01$ | $x^{(1)} = -0{,}14$<br>$x^{(2)} = 1{,}43$ |
| 24. | $x = -0{,}36$ | $x^{(1)} = -0{,}29$<br>$x^{(2)} = -0{,}68$ |
| 25. | $x = -3{,}002$ | $x^{(1)} = -4{,}817$<br>$x^{(2)} = -3{,}762$ |
| 26. | $x = 3{,}63$ | $x^{(1)} = 1{,}08$<br>$x^{(2)} = 4{,}78$ |
| 27. | $x = -13{,}18$ | $x^{(1)} = -13{,}18$<br>$x^{(2)} = -8{,}95$ |
| 28. | $x = 4{,}00$ | $x^{(1)} = 5{,}78$<br>$x^{(2)} = 4{,}00$ |
| 29. | $x = -1{,}989$ | $x^{(1)} = -1{,}564$<br>$x^{(2)} = -2{,}541$ |
| 30. | $x = 3{,}77$ | $x^{(1)} = 4{,}08$<br>$x^{(2)} = 3{,}69$ |
| 31. | $x = 2{,}82$ | $x^{(1)} = 1{,}99$<br>$x^{(2)} = 2{,}97$ |

| | | |
|---|---|---|
| 32. | $x = -7{,}62$ | $x^{(1)} = -8{,}78$<br>$x^{(2)} = -8{,}16$ |
| 33. | $x = 10{,}01$ | $x^{(1)} = 10{,}02$<br>$x^{(2)} = 8{,}37$ |
| 34. | $x = -4{,}16$ | $x^{(1)} = -3{,}99$<br>$x^{(2)} = -2{,}21$ |
| 35. | $x = -1{,}39$ | $x^{(1)} = -1{,}97$<br>$x^{(2)} = -0{,}50$ |
| 36. | $x = 12{,}721$ | $x^{(1)} = 12{,}520$<br>$x^{(2)} = 11{,}512$ |
| 37. | $x = -7{,}21$ | $x^{(1)} = -3{,}62$<br>$x^{(2)} = -8{,}12$ |
| 38. | $x = 8{,}99$ | $x^{(1)} = 10{,}17$<br>$x^{(2)} = 9{,}61$ |
| 39. | $x = -8{,}53$ | $x^{(1)} = -8{,}53$<br>$x^{(2)} = -10{,}75$ |
| 40. | $x = 0{,}33$ | $x^{(1)} = -0{,}07$<br>$x^{(2)} = 0{,}68$ |
| 41. | $x = 7{,}13$ | $x^{(1)} = 6{,}00$<br>$x^{(2)} = 8{,}02$ |
| 42. | $x = -10{,}01$ | $x^{(1)} = -7{,}83$<br>$x^{(2)} = -10{,}03$ |
| 43. | $x = 0{,}66$ | $x^{(1)} = 0{,}60$<br>$x^{(2)} = 0{,}36$ |
| 44. | $x = 0{,}19$ | $x^{(1)} = 2{,}10$<br>$x^{(2)} = -0{,}09$ |
| 45. | $x = 0{,}09$ | $x^{(1)} = -0{,}13$<br>$x^{(2)} = -0{,}50$ |
| 46. | $x = -4{,}321$ | $x^{(1)} = -5{,}282$<br>$x^{(2)} = -3{,}002$ |

| | | |
|---|---|---|
| 47. | $x = 4{,}96$ | $x^{(1)} = 6{,}17$<br>$x^{(2)} = 2{,}11$ |
| 48. | $x = -11{,}02$ | $x^{(1)} = -12{,}44$<br>$x^{(2)} = -8{,}02$ |
| 49. | $x = 4{,}06$ | $x^{(1)} = 4{,}06$<br>$x^{(2)} = 5{,}08$ |
| 50. | $x = -1{,}537$ | $x^{(1)} = -1{,}537$<br>$x^{(2)} = -2{,}001$ |
| 51. | $x = 3{,}82$ | $x^{(1)} = 3{,}55$<br>$x^{(2)} = 4{,}97$ |
| 52. | $x = 2{,}81$ | $x^{(1)} = 2{,}94$<br>$x^{(2)} = 2{,}77$ |
| 53. | $x = -7{,}37$ | $x^{(1)} = -9{,}87$<br>$x^{(2)} = -8{,}02$ |
| 54. | $x = 9{,}05$ | $x^{(1)} = 8{,}51$<br>$x^{(2)} = 9{,}77$ |
| 55. | $x = -3{,}66$ | $x^{(1)} = -4{,}91$<br>$x^{(2)} = -2{,}22$ |
| 56. | $x = -0{,}00$ | $x^{(1)} = -0{,}18$<br>$x^{(2)} = -1{,}50$ |
| 57. | $x = 12{,}132$ | $x^{(1)} = 11{,}512$<br>$x^{(2)} = 12{,}121$ |
| 58. | $x = -8{,}61$ | $x^{(1)} = -5{,}37$<br>$x^{(2)} = -8{,}73$ |
| 59. | $x = 9{,}67$ | $x^{(1)} = 10{,}68$<br>$x^{(2)} = 9{,}02$ |
| 60. | $x = -9{,}47$ | $x^{(1)} = -11{,}42$<br>$x^{(2)} = -9{,}36$ |
| 61. | $x = 0{,}52$ | $x^{(1)} = 0{,}24$<br>$x^{(2)} = 1{,}22$ |

| | | |
|---|---|---|
| 62. | $x = 7{,}88$ | $x^{(1)} = 8{,}24$<br>$x^{(2)} = 5{,}59$ |
| 63. | $x = -7{,}93$ | $x^{(1)} = -9{,}05$<br>$x^{(2)} = -7{,}91$ |
| 64. | $x = 0{,}37$ | $x^{(1)} = 0{,}69$<br>$x^{(2)} = 0{,}39$ |
| 65. | $x = -0{,}08$ | $x^{(1)} = -0{,}18$<br>$x^{(2)} = 2{,}44$ |
| 66. | $x = -0{,}55$ | $x^{(1)} = -0{,}11$<br>$x^{(2)} = -0{,}45$ |
| 67. | $x = -3{,}952$ | $x^{(1)} = -5{,}655$<br>$x^{(2)} = -3{,}251$ |
| 68. | $x = 3{,}63$ | $x^{(1)} = 2{,}70$<br>$x^{(2)} = 5{,}90$ |
| 69. | $x = -7{,}11$ | $x^{(1)} = -13{,}44$<br>$x^{(2)} = -7{,}58$ |
| 70. | $x = 5{,}60$ | $x^{(1)} = 4{,}11$<br>$x^{(2)} = 5{,}49$ |
| 71. | $x = -1{,}401$ | $x^{(1)} = -1{,}795$<br>$x^{(2)} = -2{,}121$ |
| 72. | $x = 3{,}82$ | $x^{(1)} = 4{,}82$<br>$x^{(2)} = 3{,}52$ |
| 73. | $x = 2{,}88$ | $x^{(1)} = 2{,}70$<br>$x^{(2)} = 2{,}89$ |
| 74. | $x = -7{,}42$ | $x^{(1)} = -8{,}92$<br>$x^{(2)} = -7{,}39$ |
| 75. | $x = 7{,}48$ | $x^{(1)} = 8{,}75$<br>$x^{(2)} = 9{,}29$ |

Функция для многочленов Ньютона с конечными разностями, Гаусса, Стирлинга, Бесселя: таблица значений функции $y = f(x)$ с шагом $h$ от $a$ до $b$

| № варианта | $f(x)$ | $h$ | $a$ | $b$ |
|---|---|---|---|---|
| 1. | $\sqrt[3]{x}$ | 2,0 | 0,0 | 10,0 |
| 2. | $\cos\frac{x}{3}$ | $\frac{\pi}{2}$ | 0,0 | $3\pi$ |
| 3. | $e^{x+3}$ | 0,02 | 11,5 | 11,6 |
| 4. | $\arcsin(2x)$ | 0,2 | $-0{,}5$ | 0,5 |
| 5. | $\arccos\left(\log_2\left(\frac{x}{2}\right)\right)$ | 0,5 | 1 | 4 |
| 6. | $\ln\left(\frac{2}{\sqrt{x}}\right)$ | 2 | 2 | 12 |
| 7. | $x - 2\cos(x)$ | $\frac{\pi}{6}$ | $-\frac{\pi}{6}$ | $\frac{7\pi}{6}$ |
| 8. | $x^{x-(\ln x)^2}$ | 0,2 | 1 | 2 |
| 9. | $(x - \sin x)^2 - \frac{1}{2}e^x + 1$ | 0,2 | 2 | 3 |
| 10. | $(\cos(x) - 2)^2$ | $\frac{\pi}{5}$ | 0 | $\pi$ |
| 11. | $\mathrm{tg}(\ln(x))$ | 0,5 | 1 | 4 |
| 12. | $\mathrm{arctg}(\exp x^2)$ | 0,4 | 0 | 2 |
| 13. | $\sec\left(\frac{x}{2}\right)$ | 0,4 | 0 | 2 |

| | | | | |
|---|---|---|---|---|
| 14. | $\operatorname{cosec}(2x)$ | 0,1 | 0,5 | 1 |
| 15. | $e-(\ln(x))^2$ | 1 | 1 | 5 |
| 16. | $\sqrt{1+x}-\frac{1}{\lg 2x}$ | 1 | 1 | 6 |
| 17. | $\operatorname{arctg}\left(\frac{x}{\pi}\right)$ | 1 | 0 | 6 |
| 18. | $x^{\frac{5}{\pi}}$ | 0,4 | 0 | 2 |
| 19. | $\ln\left(\frac{x}{e}\right)$ | 4 | 1 | 21 |
| 20. | $\sqrt{x^{x-1}-1}+\cos(x)$ | 0,5 | 1 | 4 |
| 21. | $-\lg(\sin^2(x))$ | 0,5 | $-6$ | $-3{,}5$ |
| 22. | $\sqrt[3]{x}$ | 2,0 | 0,0 | 10,0 |
| 23. | $\cos\frac{x}{3}$ | $\frac{\pi}{2}$ | 0,0 | $3\pi$ |
| 24. | $e^{x+3}$ | 0,02 | 11,5 | 11,6 |
| 25. | $\arcsin(2x)$ | 0,2 | $-0{,}5$ | 0,5 |
| 26. | $\arccos\left(\log_2\left(\frac{x}{2}\right)\right)$ | 0,5 | 1 | 4 |
| 27. | $\ln\left(\frac{2}{\sqrt{x}}\right)$ | 2 | 2 | 12 |
| 28. | $x-2\cos(x)$ | $\frac{\pi}{6}$ | $-\frac{\pi}{6}$ | $\frac{7\pi}{6}$ |

| | | | | |
|---|---|---|---|---|
| 29. | $x^{x-(\ln x)^2}$ | 0,2 | 1 | 2 |
| 30. | $(x-\sin x)^2-\frac{1}{2}e^x+1$ | 0,2 | 2 | 3 |
| 31. | $(\cos(x)-2)^2$ | $\frac{\pi}{5}$ | 0 | $\pi$ |
| 32. | $\mathrm{tg}(\ln(x))$ | 0,5 | 1 | 4 |
| 33. | $\mathrm{arctg}(\exp x^2)$ | 0,4 | 0 | 2 |
| 34. | $\sec\left(\frac{x}{2}\right)$ | 0,4 | 0 | 2 |
| 35. | $\mathrm{cosec}(2x)$ | 0,1 | 0,5 | 1 |
| 36. | $e-(\ln(x))^2$ | 1 | 1 | 5 |
| 37. | $\sqrt{1+x}-\frac{1}{\lg 2x}$ | 1 | 1 | 6 |
| 38. | $\mathrm{arctg}\left(\frac{x}{\pi}\right)$ | 1 | 0 | 6 |
| 39. | $x^{\frac{5}{\pi}}$ | 0,4 | 0 | 2 |
| 40. | $\ln\left(\frac{x}{e}\right)$ | 4 | 1 | 21 |
| 41. | $\sqrt{x^{x-1}-1}+\cos(x)$ | 0,5 | 1 | 4 |
| 42. | $-\lg(\sin^2(x))$ | 0,5 | $-6$ | $-3{,}5$ |
| 43. | $\sqrt[3]{x}$ | 2,0 | 0,0 | 10,0 |

| | | | | |
|---|---|---|---|---|
| 44. | $\cos\frac{x}{3}$ | $\frac{\pi}{2}$ | 0,0 | $3\pi$ |
| 45. | $e^{x+3}$ | 0,02 | 11,5 | 11,6 |
| 46. | $\arcsin(2x)$ | 0,2 | $-0{,}5$ | 0,5 |
| 47. | $\arccos\left(\log_2\left(\frac{x}{2}\right)\right)$ | 0,5 | 1 | 4 |
| 48. | $\ln\left(\frac{2}{\sqrt{x}}\right)$ | 2 | 2 | 12 |
| 49. | $x-2\cos(x)$ | $\frac{\pi}{6}$ | $-\frac{\pi}{6}$ | $\frac{7\pi}{6}$ |
| 50. | $x^{x-(\ln x)^2}$ | 0,2 | 1 | 2 |
| 51. | $(x-\sin x)^2-\frac{1}{2}e^x+1$ | 0,2 | 2 | 3 |
| 52. | $(\cos(x)-2)^2$ | $\frac{\pi}{5}$ | 0 | $\pi$ |
| 53. | $\mathrm{tg}(\ln(x))$ | 0,5 | 1 | 4 |
| 54. | $\mathrm{arctg}(\exp x^2)$ | 0,4 | 0 | 2 |
| 55. | $\sec\left(\frac{x}{2}\right)$ | 0,4 | 0 | 2 |
| 56. | $\mathrm{cosec}(2x)$ | 0,1 | 0,5 | 1 |
| 57. | $e-(\ln(x))^2$ | 1 | 1 | 5 |
| 58. | $\sqrt{1+x}-\frac{1}{\lg 2x}$ | 1 | 1 | 6 |

| | | | | |
|---|---|---|---|---|
| 59. | $\operatorname{arctg}\left(\frac{x}{\pi}\right)$ | 1 | 0 | 6 |
| 60. | $x^{\frac{5}{\pi}}$ | 0,4 | 0 | 2 |
| 61. | $\ln\left(\frac{x}{e}\right)$ | 4 | 1 | 21 |
| 62. | $\sqrt{x^{x-1}-1}+\cos(x)$ | 0,5 | 1 | 4 |
| 63. | $-\lg(\sin^2(x))$ | 0,5 | $-6$ | $-3{,}5$ |
| 64. | $\sqrt[3]{x}$ | 2,0 | 0,0 | 10,0 |
| 65. | $\cos\frac{x}{3}$ | $\frac{\pi}{2}$ | 0,0 | $3\pi$ |
| 66. | $e^{x+3}$ | 0,02 | 11,5 | 11,6 |
| 67. | $\arcsin(2x)$ | 0,2 | $-0{,}5$ | 0,5 |
| 68. | $\arccos\left(\log_2\left(\frac{x}{2}\right)\right)$ | 0,5 | 1 | 4 |
| 69. | $\ln\left(\frac{2}{\sqrt{x}}\right)$ | 2 | 2 | 12 |
| 70. | $x-2\cos(x)$ | $\frac{\pi}{6}$ | $-\frac{\pi}{6}$ | $\frac{7\pi}{6}$ |
| 71. | $x^{x-(\ln x)^2}$ | 0,2 | 1 | 2 |
| 72. | $(x-\sin x)^2-\frac{1}{2}e^x+1$ | 0,2 | 2 | 3 |
| 73. | $(\cos(x)-2)^2$ | $\frac{\pi}{5}$ | 0 | $\pi$ |

| | | | | |
|---|---|---|---|---|
| 74. | $\mathrm{tg}(\ln(x))$ | 0,5 | 1 | 4 |
| 75. | $\mathrm{arctg}(\exp x^2)$ | 0,4 | 0 | 2 |

Примечания: 1) функцию вычислять с двумя значащими цифрами после запятой в мантиссе; 2) значение $\pi$ брать $3{,}14$.

Точки интерполяции (экстраполяции) многочленами Ньютона с конечными разностями, Гаусса, Стирлинга, Бесселя

| № варианта | Многочлен Ньютона с конечными разностями | Многочлены Гаусса, Стирлинга, Бесселя |
|---|---|---|
| 1. | $x^{(1)} = -0{,}02$<br>$x^{(2)} = 10{,}01$ | $x = 4{,}7$ |
| 2. | $x^{(1)} = -\frac{\pi}{8}$<br>$x^{(2)} = 2{,}73\pi$ | $x = 1{,}53\pi$ |
| 3. | $x^{(1)} = 11{,}57$<br>$x^{(2)} = 11{,}41$ | $x = 11{,}27$ |
| 4. | $x^{(1)} = -0{,}3$<br>$x^{(2)} = 0{,}3$ | $x = 0$ |
| 5. | $x^{(1)} = 1{,}3$<br>$x^{(2)} = 3{,}85$ | $x = 2{,}3$ |
| 6. | $x^{(1)} = 0{,}9$<br>$x^{(2)} = 11{,}6$ | $x = 5{,}35$ |
| 7. | $x^{(1)} = \frac{\pi}{4}$<br>$x^{(2)} = 1{,}3\pi$ | $x = 0{,}7\pi$ |
| 8. | $x^{(1)} = 1{,}03$<br>$x^{(2)} = 1{,}9$ | $x = 1{,}51$ |
| 9. | $x^{(1)} = 2{,}1$<br>$x^{(2)} = 3{,}08$ | $x = 2{,}48$ |

| | | |
|---|---|---|
| 10. | $x^{(1)} = -0{,}2\pi$<br>$x^{(2)} = 0{,}8\pi$ | $x = 0{,}33\pi$ |
| 11. | $x^{(1)} = 0{,}8$<br>$x^{(2)} = 3{,}6$ | $x = 2{,}2$ |
| 12. | $x^{(1)} = 0{,}7$<br>$x^{(2)} = 2{,}5$ | $x = 1{,}33$ |
| 13. | $x^{(1)} = -0{,}02$<br>$x^{(2)} = 2{,}16$ | $x = 1{,}00$ |
| 14. | $x^{(1)} = 0{,}4$<br>$x^{(2)} = 0{,}99$ | $x = 0{,}75$ |
| 15. | $x^{(1)} = 0{,}77$<br>$x^{(2)} = 4{,}82$ | $x = 3{,}2$ |
| 16. | $x^{(1)} = 1{,}56$<br>$x^{(2)} = 6{,}23$ | $x = 4{,}26$ |
| 17. | $x^{(1)} = -0{,}29$<br>$x^{(2)} = 5{,}56$ | $x = 3{,}02$ |
| 18. | $x^{(1)} = -0{,}78$<br>$x^{(2)} = 1{,}43$ | $x = 0{,}50$ |
| 19. | $x^{(1)} = 2{,}15$<br>$x^{(2)} = 22{,}56$ | $x = 13{,}5$ |
| 20. | $x^{(1)} = 1{,}4$<br>$x^{(2)} = 4{,}3$ | $x = 2{,}85$ |
| 21. | $x^{(1)} = -5{,}34$<br>$x^{(2)} = -3{,}98$ | $x = -4{,}8$ |
| 22. | $x^{(1)} = -0{,}06$<br>$x^{(2)} = 10{,}09$ | $x = 4{,}8$ |
| 23. | $x^{(1)} = -\frac{\pi}{8}$<br>$x^{(2)} = 2{,}77\pi$ | $x = 1{,}52\pi$ |

| | | |
|---|---|---|
| 24. | $x^{(1)} = 11{,}59$<br>$x^{(2)} = 11{,}4$ | $x = 11{,}25$ |
| 25. | $x^{(1)} = -0{,}32$<br>$x^{(2)} = 0{,}32$ | $x = -0{,}1$ |
| 26. | $x^{(1)} = 1{,}11$<br>$x^{(2)} = 3{,}82$ | $x = 2{,}2$ |
| 27. | $x^{(1)} = 0{,}92$<br>$x^{(2)} = 11{,}63$ | $x = 5{,}33$ |
| 28. | $x^{(1)} = \frac{\pi}{4}$<br>$x^{(2)} = 1{,}33\pi$ | $x = 0{,}8\pi$ |
| 29. | $x^{(1)} = 1{,}01$<br>$x^{(2)} = 1{,}92$ | $x = 1{,}57$ |
| 30. | $x^{(1)} = 2{,}12$<br>$x^{(2)} = 3{,}04$ | $x = 2{,}44$ |
| 31. | $x^{(1)} = -0{,}02\pi$<br>$x^{(2)} = 0{,}79\pi$ | $x = 0{,}38\pi$ |
| 32. | $x^{(1)} = 0{,}79$<br>$x^{(2)} = 3{,}62$ | $x = 2{,}1$ |
| 33. | $x^{(1)} = 0{,}72$<br>$x^{(2)} = 2{,}47$ | $x = 1{,}30$ |
| 34. | $x^{(1)} = -0{,}03$<br>$x^{(2)} = 2{,}11$ | $x = 1{,}08$ |
| 35. | $x^{(1)} = 0{,}38$<br>$x^{(2)} = 0{,}92$ | $x = 0{,}76$ |
| 36. | $x^{(1)} = 0{,}79$<br>$x^{(2)} = 4{,}80$ | $x = 3{,}22$ |
| 37. | $x^{(1)} = 1{,}52$<br>$x^{(2)} = 6{,}25$ | $x = 4{,}29$ |

| | | |
|---|---|---|
| 38. | $x^{(1)} = -0{,}30$<br>$x^{(2)} = 5{,}52$ | $x = 3{,}06$ |
| 39. | $x^{(1)} = -0{,}73$<br>$x^{(2)} = 1{,}40$ | $x = 0{,}54$ |
| 40. | $x^{(1)} = 2{,}17$<br>$x^{(2)} = 22{,}52$ | $x = 13{,}7$ |
| 41. | $x^{(1)} = 1{,}37$<br>$x^{(2)} = 4{,}31$ | $x = 2{,}82$ |
| 42. | $x^{(1)} = -5{,}31$<br>$x^{(2)} = -3{,}94$ | $x = -4{,}81$ |
| 43. | $x^{(1)} = -0{,}03$<br>$x^{(2)} = 10{,}12$ | $x = 4{,}82$ |
| 44. | $x^{(1)} = -\frac{\pi}{8}$<br>$x^{(2)} = 2{,}72\pi$ | $x = 1{,}54\pi$ |
| 45. | $x^{(1)} = 11{,}61$<br>$x^{(2)} = 11{,}38$ | $x = 11{,}27$ |
| 46. | $x^{(1)} = -0{,}28$<br>$x^{(2)} = 0{,}28$ | $x = -0{,}02$ |
| 47. | $x^{(1)} = 1{,}32$<br>$x^{(2)} = 3{,}84$ | $x = 2{,}2$ |
| 48. | $x^{(1)} = 0{,}91$<br>$x^{(2)} = 11{,}54$ | $x = 5{,}35$ |
| 49. | $x^{(1)} = \frac{\pi}{4}$<br>$x^{(2)} = 1{,}4\pi$ | $x = 0{,}8\pi$ |
| 50. | $x^{(1)} = 1{,}01$<br>$x^{(2)} = 1{,}93$ | $x = 1{,}56$ |
| 51. | $x^{(1)} = 2{,}12$<br>$x^{(2)} = 3{,}02$ | $x = 2{,}51$ |

| | | |
|---|---|---|
| 52. | $x^{(1)} = 0{,}03\pi$<br>$x^{(2)} = 0{,}8\pi$ | $x = 0{,}32\pi$ |
| 53. | $x^{(1)} = 0{,}7$<br>$x^{(2)} = 3{,}7$ | $x = 2{,}23$ |
| 54. | $x^{(1)} = 0{,}6$<br>$x^{(2)} = 2{,}6$ | $x = 1{,}31$ |
| 55. | $x^{(1)} = -0{,}01$<br>$x^{(2)} = 2{,}19$ | $x = 0{,}98$ |
| 56. | $x^{(1)} = 0{,}42$<br>$x^{(2)} = 0{,}94$ | $x = 0{,}77$ |
| 57. | $x^{(1)} = 0{,}75$<br>$x^{(2)} = 4{,}84$ | $x = 3{,}15$ |
| 58. | $x^{(1)} = 1{,}54$<br>$x^{(2)} = 6{,}28$ | $x = 4{,}21$ |
| 59. | $x^{(1)} = -0{,}28$<br>$x^{(2)} = 5{,}52$ | $x = 3{,}00$ |
| 60. | $x^{(1)} = -0{,}74$<br>$x^{(2)} = 1{,}41$ | $x = 0{,}53$ |
| 61. | $x^{(1)} = 2{,}15$<br>$x^{(2)} = 22{,}56$ | $x = 13{,}47$ |
| 62. | $x^{(1)} = 1{,}42$<br>$x^{(2)} = 4{,}27$ | $x = 2{,}82$ |
| 63. | $x^{(1)} = -5{,}30$<br>$x^{(2)} = -4{,}02$ | $x = -4{,}8$ |
| 64. | $x^{(1)} = -0{,}08$<br>$x^{(2)} = 10{,}12$ | $x = 4{,}8$ |
| 65. | $x^{(1)} = -\frac{\pi}{8}$<br>$x^{(2)} = 2{,}73\pi$ | $x = 1{,}55\pi$ |

| | | |
|---|---|---|
| 66. | $x^{(1)} = 11{,}63$<br>$x^{(2)} = 11{,}42$ | $x = 11{,}21$ |
| 67. | $x^{(1)} = -0{,}33$<br>$x^{(2)} = 0{,}33$ | $x = 0{,}08$ |
| 68. | $x^{(1)} = 1{,}28$<br>$x^{(2)} = 3{,}82$ | $x = 2{,}26$ |
| 69. | $x^{(1)} = 0{,}8$<br>$x^{(2)} = 11{,}7$ | $x = 5{,}33$ |
| 70. | $x^{(1)} = \frac{\pi}{4}$<br>$x^{(2)} = 1{,}27\pi$ | $x = 0{,}72\pi$ |
| 71. | $x^{(1)} = 1{,}01$<br>$x^{(2)} = 1{,}92$ | $x = 1{,}54$ |
| 72. | $x^{(1)} = 2{,}07$<br>$x^{(2)} = 3{,}04$ | $x = 2{,}52$ |
| 73. | $x^{(1)} = -0{,}85$<br>$x^{(2)} = 0{,}8\pi$ | $x = 0{,}33\pi$ |
| 74. | $x^{(1)} = 0{,}8$<br>$x^{(2)} = 3{,}6$ | $x = 2{,}2$ |
| 75. | $x^{(1)} = 0{,}7$<br>$x^{(2)} = 2{,}5$ | $x = 1{,}33$ |